

Acknowledgement

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Abstract

Bedridden patients who are unable to change their body position for long periods are at high risk of developing pressure ulcers, commonly known as bed sores. Continuous manual monitoring by caregivers to prevent this is physically demanding, difficult to sustain around the clock, and prone to delay, particularly in home-care settings and understaffed healthcare facilities. This report presents the design and proposed methodology of the IoT-Based Smart Patient Turning and Bed Monitoring System, an Internet of Things (IoT) based assistive device intended to continuously monitor a bedridden patient's body pressure, position duration, and the temperature and humidity conditions of the bedding environment. The proposed system is built around an ESP32 Wi-Fi-enabled microcontroller that reads pressure data from a Force Sensitive Resistor (FSR), together with ambient temperature and humidity data from a DHT22 sensor. If the sensed pressure pattern indicates that the patient has remained in the same position beyond a predefined threshold duration (for example, two to three hours), the system is designed to trigger a local buzzer alert and to push a real-time notification to a caregiver's mobile device over Wi-Fi. Sensor data is intended to be logged to a cloud platform, such as Blynk or Firebase, enabling doctors or caregivers to review the patient's condition remotely. As an optional advanced feature, the design also allows for a servo motor or linear actuator to slightly tilt the bed automatically in response to a prolonged-pressure alert, further reducing the burden of manual repositioning. This report details the idea, feature set, methodology, hardware and software components, proposed system architecture, working principle, evaluation plan, expected outcomes, advantages, limitations, and future work associated with the proposed system, situating it within the existing body of literature on IoT-based pressure ulcer prevention and smart-bed monitoring systems.

Keywords

Internet of Things (IoT); Pressure Ulcer Prevention; Bed Sore Monitoring; Smart Bed; ESP32; Force Sensitive Resistor (FSR); Patient Position Monitoring; Remote Patient Monitoring; Assistive Healthcare Technology

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CHAPTER 1

1. Introduction

1.1 Background

Pressure ulcers, also referred to as bed sores or decubitus ulcers, are localized areas of skin and underlying tissue damage that develop as a result of sustained pressure, most commonly over bony prominences such as the shoulder blades, hips, sacrum, and heels. They occur when a patient is unable to change position frequently enough to relieve pressure on a given area, restricting blood flow to the skin and underlying tissue over time. Bedridden patients – including those recovering from surgery, individuals with spinal cord injuries, and elderly patients with limited mobility – are especially vulnerable, since they often depend entirely on caregivers to reposition them at regular intervals.

In hospital wards, nursing homes, and home-care settings alike, the standard practice for preventing pressure ulcers is manual repositioning of the patient every two to three hours. This practice, while effective in principle, places a significant and continuous burden on caregivers, who must track elapsed time for each patient and physically reposition them regardless of workload, staffing levels, or time of day. In practice, this monitoring is frequently delayed or missed altogether, particularly overnight or in facilities with a high patient-to-caregiver ratio, allowing pressure to build up on a single area for longer than is medically recommended.

The growth of low-cost embedded systems, wireless sensor modules, and the Internet of Things (IoT) has opened new possibilities for augmenting this manual process with automated, continuous sensing and alerting. Rather than relying solely on a caregiver’s memory and availability, an IoT-based system can continuously track how long pressure has been sustained on a given area and proactively alert a caregiver – or, in an extended configuration, adjust the bed itself – before a dangerous level of sustained pressure is reached.

1.2 Motivation

The motivation for this project stems from the everyday difficulty of providing truly continuous position monitoring for bedridden patients, whether in a hospital, a nursing facility, or at home. Family caregivers in particular often lack the training, the physical stamina, or simply the ability to stay awake and attentive around the clock to track how long a patient has remained in one position. Existing commercial pressure-monitoring solutions, where they exist, tend to be expensive, hospital-grade equipment that is out of reach for home-care use, and they rarely combine pressure sensing with environmental monitoring, mobile alerting, and cloud-based remote review in a single affordable package.

This project is motivated by the goal of designing a low-cost, extensible IoT platform – built from readily available components such as the ESP32, force-sensitive resistors, and a DHT22 sensor – that can give caregivers a continuous, automated safety net for pressure-ulcer prevention without requiring expensive, proprietary hardware.

1.3 Problem Statement

Bedridden patients face three interrelated risks that the traditional practice of scheduled manual repositioning does not reliably address. First, sustained pressure on a single body area for longer than the medically recommended interval is a primary driver of pressure ulcer formation, yet there is typically no automated way to detect exactly how long a patient has remained in one position. Second, prolonged skin moisture and elevated temperature at the patient-mattress interface – caused by sweating or poor ventilation – are known contributing factors to skin breakdown, and these environmental conditions are rarely monitored at all under manual care routines. Third, caregivers have no automated, real-time notification mechanism to alert them the moment a position-duration threshold is crossed, meaning that the responsibility for timely repositioning rests entirely on manual vigilance. The IoT-Based Smart Patient Turning and Bed Monitoring System proposed in this report is designed to address these three problems through continuous pressure and position-duration monitoring, integrated temperature and humidity sensing, and automated local and remote (mobile/cloud) alerting, with an optional automated bed-tilt mechanism as a further extension.

2. Literature Review

2.1 Existing Smart Bed and Pressure-Monitoring Systems

A growing body of research has explored the use of embedded sensing to detect and prevent pressure ulcers in bedridden and immobile patients. Hong [1] proposed a smart care bed for elderly patients with impaired mobility in which multiple pressure sensors are deployed beneath the mattress cover, with the patient’s body divided into vertical and horizontal zones so that pressure on individual body regions can be tracked independently; the system alerts caregivers through a mobile push-notification service once pressure on a given zone has persisted beyond a set duration. Hudec et al. [2] presented a smart IoT system built around a novel textile pressure sensor matrix, in which sixty-four pressure sensors woven into a mattress topper are read over an MQTT-based IoT pipeline and classified by a neural network into one of four lying postures, reminding care staff when a change of position is required. Memari et al. [4] specifically addressed the design and development of a pressure-monitoring system aimed at preventing bedsores in immobile patients through continuous sensing of pressure at the patient-mattress interface.

2.2 Related Works

Beyond pressure sensing alone, several studies have investigated closely related aspects of bed-based patient monitoring. Awais et al. [3] proposed an IoT-based bed-egress alerting system that uses wearable accelerometer and RFID-based sensing to detect when an elderly or at-risk patient leaves the bed unassisted, generating a timely alert to healthcare staff; although focused on bed-exit detection rather than pressure ulcer prevention, this work demonstrates the broader feasibility of low-latency IoT alerting in bed-based patient care. Abdelmoghith et al. [6] proposed an IoT-based healthcare monitoring system that specifically targets bedsores prevention by combining body-temperature sensing, skin-humidity sensing, and body-immobility-duration tracking – a combination of sensing modalities closely aligned with the approach adopted in this report. More recently, Gomathi Durga et al. [5] described an IoT-based “MediBed” system for elderly patients that combines force-sensitive-resistor pressure sensing with an ESP32 microcontroller, actuator-driven automatic bed tilting, and Firebase-based cloud connectivity for caregiver dashboards, closely mirroring the optional automatic-tilt extension considered in this proposal.

2.3 Existing Technologies

The core technologies relevant to this domain include resistive force-sensing elements such as Force Sensitive Resistors (FSR) and strain-gauge load cells (typically paired with a dedicated instrumentation amplifier such as the HX711 24-bit analogue-to-digital converter) for pressure and weight sensing; capacitive/thermistor-based digital sensors such as the DHT22 for combined temperature and humidity measurement; low-power Wi-Fi-enabled microcon-

troller platforms such as the ESP32 for sensor data acquisition and wireless communication; and cloud-connected IoT dashboard platforms such as Blynk and Google Firebase for remote data storage, visualization, and mobile push notifications. A broader review by Abdulmalek et al. [7] surveys the wider landscape of IoT-based healthcare monitoring systems, situating pressure-ulcer-focused devices such as the one proposed here within the larger trend of connected, sensor-driven remote patient monitoring.

2.4 Research Gap

Across the systems reviewed above, a recurring pattern emerges: most existing academic prototypes focus on a single capability in isolation – either pressure/posture sensing [1, 2, 4], bed-exit/fall alerting [3], or environmental sensing – with relatively few systems combining continuous pressure-duration tracking, environmental (temperature/humidity) monitoring, local audible alerting, real-time mobile notification, and cloud-based remote review within a single low-cost, self-contained device. Where automatic bed actuation is included, as in the MediBed system [5], it is a recent and still-emerging capability rather than an established standard feature. The IoT-Based Smart Patient Turning and Bed Monitoring System proposed in this report is designed to bring several of these capabilities – pressure and position-duration monitoring, environmental sensing, local and mobile alerting, cloud logging, and an optional automated tilt mechanism – together in a single affordable, extensible platform built around the ESP32.

2.5 Comparison of Existing Systems

System / Approach	Pressure Sensing	Auto Bed Actuation	Environment Sensing	Cloud / Remote Access	Caregiver Alert
Traditional manual repositioning	No	No	No	No	No
Smart Care Bed, Hong [1] (2018)	Yes	No	No	Yes (push)	Yes
Textile IoT Bed, Hudec et al. [2] (2021)	Yes	No	No	Yes (MQTT/server)	Yes
Bed-Egress Alerting, Awais et al. [3] (2019)	No (wearable)	No	No	Partial	Yes
MediBed, Gomathi Durga et al. [5] (2025)	Yes	Yes	No	Yes (Firebase)	Yes
Proposed System (this report)	Yes	Optional	Yes	Yes (Blynk/Firebase)	Yes (buzzer + mobile)

Table 2.1 Comparison of Existing IoT-Based Pressure/Bed Monitoring Systems

CHAPTER 3

3. Objectives

3.1 General Objective

The general objective of this project is to design a low-cost, IoT-enabled smart bed-monitoring system that reduces the risk of pressure ulcers in bedridden patients by continuously tracking body pressure and position duration, monitoring the temperature and humidity conditions of the bedding environment, and automatically alerting caregivers – locally and remotely – when a patient requires repositioning.

3.2 Specific Objectives

- To design a mattress-integrated hardware platform around the ESP32 microcontroller that reads body-pressure data from FSR sensors.
- To implement a time-based position-tracking routine that detects when a patient has remained in the same position for longer than a predefined threshold (for example, two to three hours).
- To integrate a DHT22 sensor for continuous monitoring of ambient temperature and humidity around the patient, in order to detect conditions associated with prolonged moisture and sweating.
- To implement a dual-channel alert mechanism consisting of a local buzzer alert and a Wi-Fi-based real-time mobile notification to the caregiver.
- To integrate the system with a cloud IoT platform (Blynk or Firebase) so that sensor data and alert history can be reviewed remotely by caregivers or doctors.
- To explore an optional automatic bed-tilt mechanism, using a servo motor or linear actuator, that can gently adjust bed angle in response to a sustained-pressure alert.
- To plan an evaluation methodology for assessing the prototype's pressure-detection responsiveness, environmental-sensing accuracy, and alert-delivery latency once implemented.

4. Proposed System

4.1 System Overview

The IoT-Based Smart Patient Turning and Bed Monitoring System is proposed as an integrated, mattress-and-bedside-mounted assistive device built around an ESP32 microcontroller acting as the central processing and communication hub. The system is organized into four functional subsystems intended to operate concurrently under the control of the main firmware loop: a sensing subsystem (body-pressure sensing via FSR, and environmental sensing via DHT22), a decision subsystem (threshold-based position-duration tracking implemented on the ESP32), an alerting subsystem (local buzzer plus Wi-Fi-based mobile notification), and a cloud-connectivity subsystem (data logging and remote monitoring via Blynk or Firebase). An optional fifth subsystem – automated bed-tilt actuation – extends the design for cases where gentle, automatic pressure redistribution is desired in addition to caregiver alerting.

The proposed design favours modularity: each sensor communicates with the ESP32 over a standard interface (an analogue ADC input for the FSR and a single-wire digital interface for the DHT22), allowing individual components to be added, replaced, or omitted – for example, adding the actuator in an extended configuration – without redesigning the overall system.

4.2 System Architecture

At the architectural level, the proposed system can be viewed as three layers. The *sensing layer* consists of the FSR pressure sensors and the DHT22 temperature and humidity sensor, which continuously sample the pressure at the patient-mattress interface and the ambient bedding conditions. The *processing and decision layer*, implemented on the ESP32, reads and timestamps these sensor values, tracks how long a given pressure condition has persisted, and determines whether a position-duration alert should be issued. The *output and communication layer* delivers a local audible alert through the buzzer (and a local visual status indication via an LED), pushes a real-time notification to the caregiver’s mobile application over Wi-Fi, and uploads sensor data to the chosen cloud platform (Blynk or Firebase) for remote review; in the extended configuration, this layer also drives the servo motor or linear actuator responsible for automatic bed-tilt adjustment. Figure 4.1 illustrates this proposed layered architecture and the intended flow of data between subsystems.

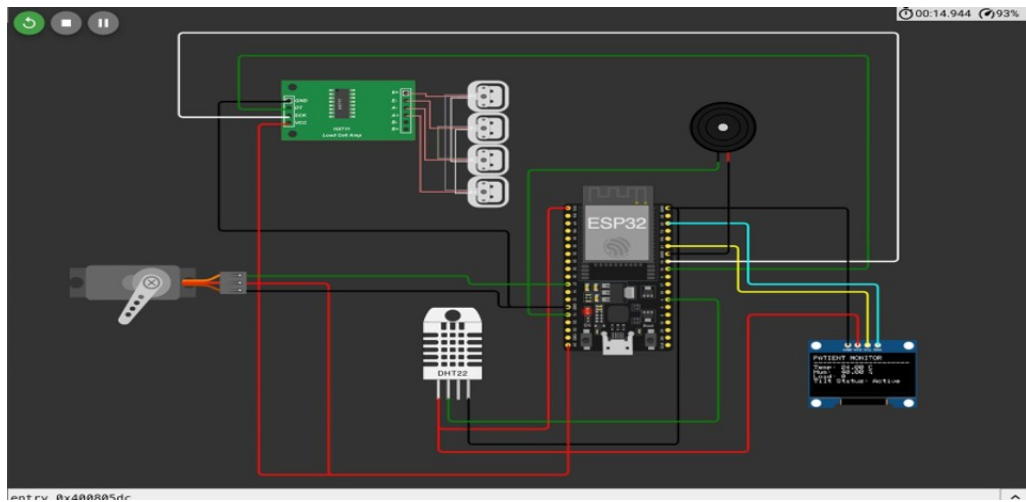


Figure 4.1 System Architecture of the IoT-Based Smart Patient Turning and Bed Monitoring System

CHAPTER 5

5. Hardware Components

This chapter describes the hardware components proposed for the IoT-Based Smart Patient Turning and Bed Monitoring System, along with the functional role each component is intended to play within the overall design.

5.1 ESP32 (Main Controller)

The ESP32 is proposed as the central microcontroller of the system. As a dual-core, Wi-Fi- and Bluetooth-enabled development board with ample GPIO, ADC, and I2C/UART interfaces, it is well suited to concurrently sampling multiple sensors, running the position-duration timing logic, driving the local buzzer, the status LED, and the optional actuator output, and maintaining a Wi-Fi connection for mobile notifications and cloud data upload.

5.2 FSR (Force Sensitive Resistor)

Body pressure at the patient-mattress interface is proposed to be sensed using Force Sensitive Resistors (FSR) placed beneath the mattress at pressure-prone body zones (for example, the shoulders, hips, and heels), consistent with the zone-based placement strategy used in related smart-bed research [1]. An FSR's electrical resistance decreases as applied force increases, allowing pressure to be read as an analogue voltage through a simple voltage-divider circuit connected to the ESP32's ADC input. For the initial testing prototype, a single FSR is used to validate the pressure-sensing and position-duration logic, with the multi-zone arrangement described above representing the intended full-scale deployment.

5.3 DHT22 Temperature and Humidity Sensor

The DHT22 (also known as the AM2302) is proposed for monitoring the ambient temperature and relative humidity around the patient. It provides a calibrated digital output over a single-wire interface, with a temperature measurement range of -40°C to 80°C at an accuracy of approximately $\pm 0.5^{\circ}\text{C}$, and a relative-humidity measurement range of 0% to 100%RH at an accuracy of approximately $\pm 2\%\text{RH}$, sampling at intervals of about two seconds. This makes it suitable for detecting the kind of localized warmth and moisture build-up (from sweating or poor ventilation) that is associated with an elevated risk of skin breakdown.

5.4 Buzzer

A compact, active buzzer is proposed as the system's local audible-alert device, providing an immediate on-site sound alert whenever a position-duration threshold is crossed, independent of whether a Wi-Fi or mobile connection is currently available.

5.5 LED Indicator

A status indicator LED provides simple local visual feedback at the bedside – for example, lighting up when a position-duration alert is active – for caregivers who are physically present in the room. It is driven directly from a digital GPIO pin on the ESP32 through a current-limiting resistor.

5.6 Servo Motor or Linear Actuator (Optional)

For the optional automatic bed-tilt feature, a small hobby servo motor (for prototyping and small-scale demonstration) or a 12 V linear actuator (for a full-scale bed frame) is proposed to gently adjust bed angle in response to a sustained-pressure alert, redistributing pressure away from the affected zone.

5.7 Power Supply

The system is proposed to be powered by a regulated 5 V supply for the ESP32 and sensor/logic-level modules, with a separate 12 V supply rail available for the optional linear actuator where the extended configuration is used.

5.8 IoT Platform (Blynk or Firebase)

Blynk or Google Firebase is proposed as the cloud connectivity layer, providing a ready-made dashboard and mobile-app framework for visualizing sensor data, storing a history of pressure and environmental readings, and delivering push notifications to a caregiver's phone whenever a position-duration alert is triggered.

5.9 Summary of Hardware Components

Component	Function	Interface
ESP32	Central microcontroller, Wi-Fi connectivity, sensor fusion and decision logic	—
FSR (Force Sensitive Resistor)	Body-pressure sensing at the patient-mattress interface	Analogue (ADC)
DHT22	Ambient temperature and humidity sensing	Digital (1-wire)
Buzzer	Local audible alert	Digital GPIO
LED Indicator	Local visual status/alert indication	Digital GPIO
Servo / Linear Actuator (optional)	Automatic bed-tilt adjustment	PWM
Power Supply (5 V / 12 V)	System and actuator power source	—
IoT Platform (Blynk / Firebase)	Cloud data logging, dashboard, mobile alerts	Wi-Fi (HTTP/MQTT)

Table 5.1 Summary of Hardware Components

CHAPTER 6

6. Software Technologies

6.1 Arduino IDE

The Arduino Integrated Development Environment is proposed as the primary tool for writing, compiling, and uploading firmware to the ESP32 microcontroller, taking advantage of its extensive library ecosystem for interfacing with the FSR, DHT22, buzzer, and Wi-Fi/cloud modules used in this project.

6.2 ESP32 Framework

The ESP32 board-support package for the Arduino framework exposes the chip's Wi-Fi radio, ADC, I2C, UART, and PWM peripherals through a familiar Arduino-style API, supporting the concurrent, non-blocking sensor-polling and communication logic required by the proposed system.

6.3 Blynk / Firebase Cloud Platform

Blynk and Firebase are both proposed as candidate cloud back-ends for the system. Blynk offers a low-code, mobile-app-first approach well suited to rapid prototyping of sensor dashboards and push notifications, while Firebase's Realtime Database offers more flexible, developer-configurable data storage and a foundation for a custom caregiver mobile or web dashboard. The final choice between the two is expected to be made during implementation based on the desired balance between development speed and customizability.

6.4 Threshold-Based Timer Logic

The core decision-making logic of the firmware is proposed to be a threshold-based timer routine: for each monitored pressure zone, the ESP32 tracks the elapsed time since the zone's pressure reading last changed significantly, and compares this elapsed time against a configurable threshold (for example, two to three hours) to determine whether a position-duration alert should be triggered.

6.5 Mobile Notification System

A Wi-Fi-based mobile notification mechanism – delivered through the chosen cloud platform's push-notification service (Blynk's mobile app or a Firebase Cloud Messaging integration) – is proposed to inform the caregiver's smartphone in real time whenever a position-duration alert is triggered locally.

CHAPTER 7

7. Working Principle

This chapter describes the intended step-by-step operating logic of the proposed system's core functions: pressure and position monitoring, the time-based alert mechanism, environmental monitoring, cloud data logging, and the optional automatic bed-tilt mechanism. All of these functions are designed to run concurrently within the main ESP32 firmware loop.

7.1 Pressure and Position Monitoring

The pressure-monitoring routine is designed to operate continuously whenever the device is powered on. The ESP32 is intended to periodically sample the FSR sensors placed beneath the mattress at each monitored body zone, converting each reading into a calibrated pressure value. The firmware is designed to track, for each zone, how long the pressure reading has remained within the same general range, treating a sustained, unchanging pressure pattern as an indication that the patient has not changed position.

7.2 Time-Based Position Alert Mechanism

Whenever the elapsed time for a given pressure zone exceeds the predefined position-duration threshold (for example, two to three hours), the firmware is designed to immediately trigger the alert sequence: activating the local buzzer, and sending a Wi-Fi-based notification to the caregiver's mobile application indicating which zone requires attention. Once the patient's position changes – detected as a significant shift in the pressure pattern across the monitored zones – the elapsed-time timer for the previously active zone is designed to reset.

7.3 Environmental Monitoring

In parallel with pressure sensing, the DHT22 sensor is designed to be polled at regular intervals (consistent with its approximately two-second minimum sampling interval) to track ambient temperature and humidity around the patient. If humidity readings indicate sustained dampness, or temperature readings indicate an elevated risk of overheating and sweating, this information is designed to be logged alongside the pressure data and made available to caregivers through the cloud dashboard.

7.4 Cloud Data Logging and Remote Monitoring

Sensor readings and alert events are designed to be periodically uploaded from the ESP32 to the chosen cloud platform (Blynk or Firebase) over Wi-Fi, allowing doctors or caregivers to remotely review the patient's pressure history, environmental conditions, and past alerts from a mobile or web dashboard, without needing to be physically present at the bedside.

7.5 Automatic Bed Tilt (Optional Advanced Feature)

In the extended configuration, a sustained-pressure alert is designed to also trigger the optional actuation subsystem: the ESP32 drives the servo motor or linear actuator to gently tilt the affected section of the bed, redistributing pressure away from the zone in question as a first-line, automatic mitigation measure that complements – rather than replaces – caregiver-performed repositioning.

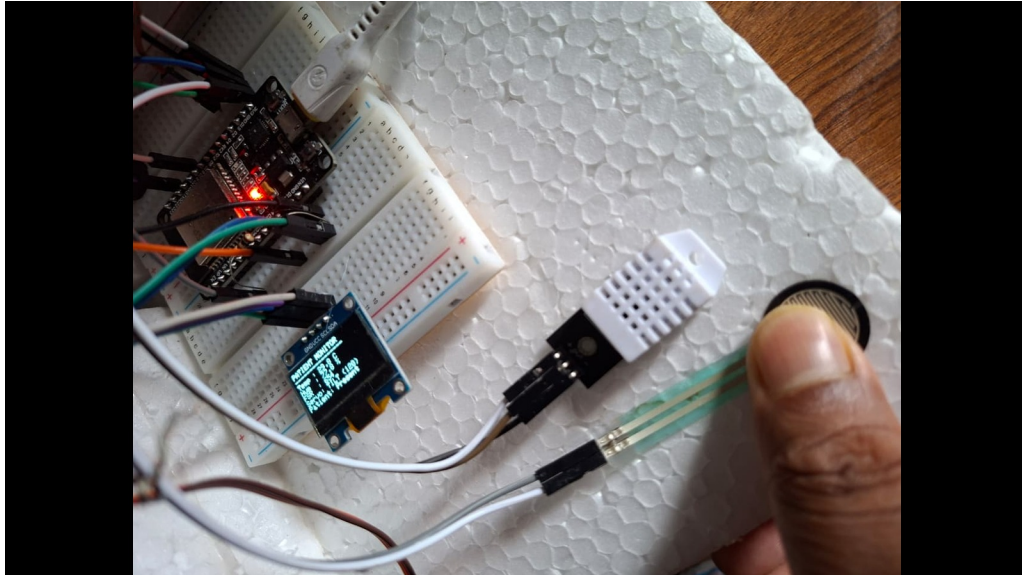


Figure 7.1 Pressure and Position Monitoring Process Flow

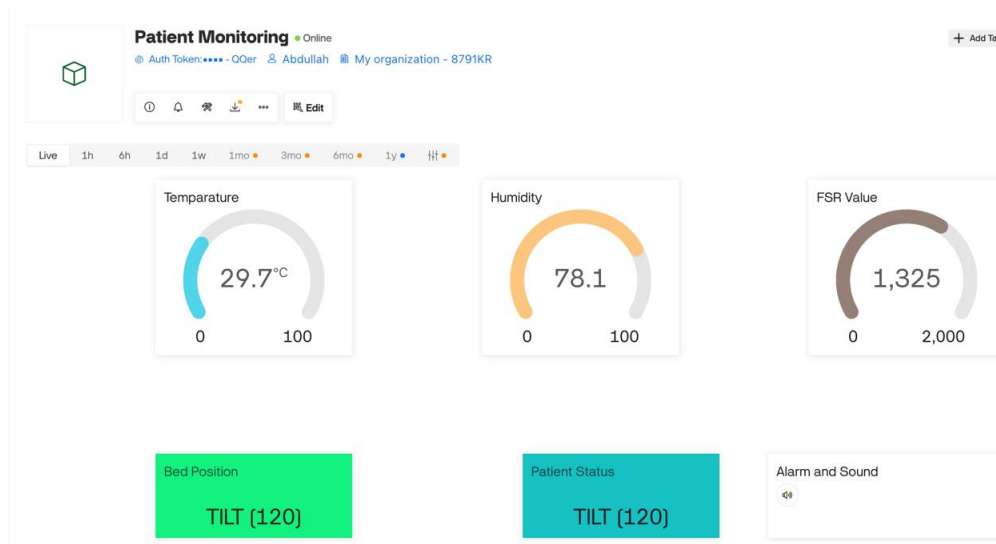


Figure 7.2 Cloud Alerting and Remote Monitoring Process Flow

CHAPTER 8

8. System Block Diagram

Figure 8.1 presents the overall proposed system block diagram, illustrating how the sensing, decision, alerting, and cloud-connectivity subsystems described in Chapter 7 are intended to be coordinated within a single firmware execution loop on the ESP32. After system initialization and sensor calibration, the main loop is designed to continuously poll the FSR pressure sensors and the DHT22 environmental sensor, evaluate the position-duration timers for each monitored zone, and service the buzzer, mobile-notification, cloud-upload, and (where enabled) actuator outputs, before returning to the start of the loop.

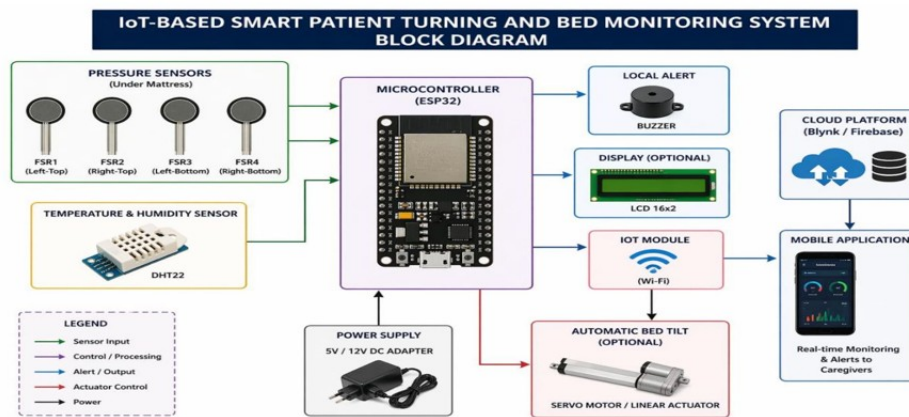


Figure 8.1 Overall System Block Diagram

CHAPTER 9

9. Circuit Diagram

Figure 9.1 shows the proposed circuit diagram of the IoT-Based Smart Patient Turning and Bed Monitoring System, detailing the intended electrical connections between the ESP32 and each peripheral module. The FSR sensors are proposed to be connected to the ESP32's analogue (ADC) input pins through simple voltage-divider circuits, The DHT22 sensor is proposed to connect to a single digital GPIO pin with a pull-up resistor on the data line, consistent with its standard single-wire interface. The buzzer is proposed to be driven from a digital GPIO pin, the status LED is proposed to be connected to a digital GPIO pin through a current-limiting resistor, and the optional servo motor or linear actuator is proposed to be driven from a PWM-capable GPIO pin (with a separate, appropriately rated driver stage for a 12 V linear actuator). Power for the 3.3 V/5 V logic-level modules is intended to be drawn from a regulated 5 V supply, while any actuator requiring a higher voltage or current is intended to be powered from a separate, appropriately rated 12 V rail.

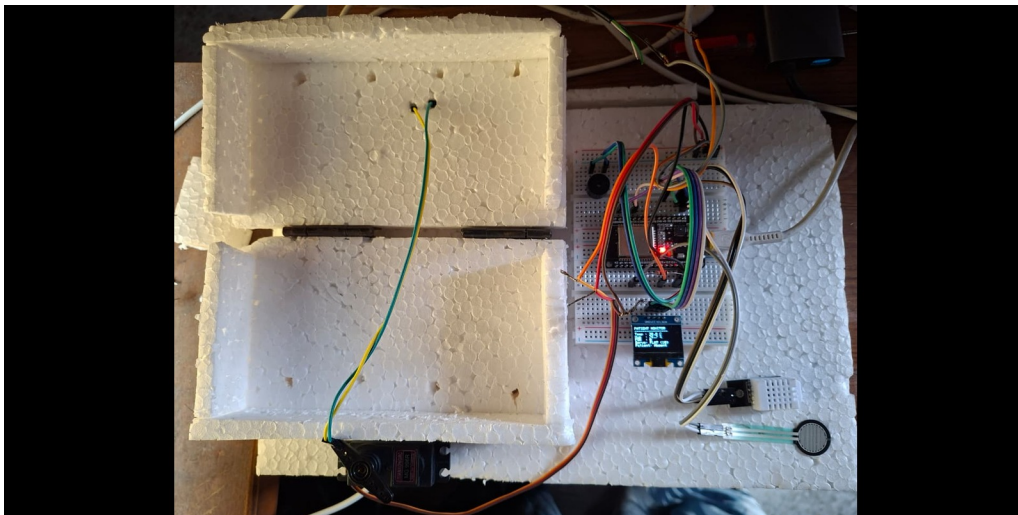


Figure 9.1 Circuit Diagram of the IoT-Based Smart Patient Turning and Bed Monitoring System

CHAPTER 10

10. Sensor Placement and Physical Design

This chapter describes the proposed physical layout of the sensing hardware within the mattress and bed frame, and the intended placement of the bedside electronics enclosure.

Consistent with the zone-based sensing approach used in related smart-bed research [1], the FSR sensors are proposed to be placed beneath the mattress cover at the body regions most commonly associated with pressure-ulcer formation – namely the shoulder/scapula region, the hips/sacrum, and the heels. The DHT22 sensor is proposed to be mounted at a point close to the patient’s torso, beneath the top bedsheet, where it can sense localized warmth and humidity without direct, uncomfortable contact with the skin. The ESP32, buzzer, and status LED are proposed to be housed in a compact enclosure mounted on or near the bed frame, within easy reach for maintenance and visual inspection, while remaining unobtrusive to the patient. Where the actuator is included, the servo motor or linear actuator is proposed to be mounted beneath the mattress section corresponding to the monitored pressure zone, allowing it to gently lift or tilt that section of the bed when triggered.

CHAPTER 11

11. Implementation

This chapter describes how the IoT-Based Smart Patient Turning and Bed Monitoring System was implemented as a working demonstration prototype, following the methodology described in Chapter 4. The work was carried out in four stages: hardware assembly and sensor calibration, firmware development, cloud and mobile integration, and testing and validation.

11.1 Phase 1: Hardware Assembly and Sensor Calibration

Each sensor module (the FSR pressure sensor and the DHT22) was first tested individually on a breadboard to verify correct wiring and baseline sensor readings. The pressure-sensing channel was then calibrated against known reference weights, after which the components were assembled together with the ESP32, buzzer, and power supply beneath the mattress and within the bedside enclosure.

11.2 Phase 2: Firmware Development

The firmware was developed using a modular approach, with each sensor and communication function encapsulated in its own routine – for example, a `readPressure()` routine for the FSR sensing, a `readEnvironment()` routine for the DHT22, and a `sendAlert()` routine for the buzzer and mobile-notification logic – so that each subsystem could be developed and tested independently before being combined in the main control loop. The main loop uses non-blocking, timer-based polling (rather than blocking delays) so that pressure sensing is not degraded while the system waits on slower operations such as Wi-Fi reconnection or cloud upload.

11.3 Phase 3: Cloud and Mobile Integration

Once the local sensing and alerting logic was working, the ESP32 firmware was connected to the Blynk cloud platform, and the data-upload routine, the caregiver dashboard, and the mobile push-notification pathway were configured so that sensor readings and position-duration alerts could be monitored remotely.

11.4 Phase 4: Testing and Validation

Finally, the assembled prototype was tested on the bench to confirm the end-to-end behaviour of the system: that sustained pressure on the sensor triggers the position-duration alert once the configured threshold is reached, and that the buzzer, mobile notification, and cloud update respond accordingly. The evaluation criteria and the results observed during this testing are described in Chapter 12.

CHAPTER 12

12. Evaluation and Results

This chapter presents the results obtained from the demonstration prototype. The assembled system was run and its live sensor and actuator status was logged over the ESP32 serial monitor. A representative sample of the captured output is reproduced below, and the observed values are summarized in Table 12.1. The discussion that follows is based directly on this run.

Sample serial-monitor output from the prototype:

```
Temp: 30.00C Hum: 77.20% FSR: 858 Servo: Tilt(120) Patient: Present
Temp: 30.10C Hum: 77.40% FSR: 829 Servo: Tilt(120) Patient: Present
Returning Bed Flat (10 degrees)
Temp: 30.10C Hum: 77.40% FSR: 841 Servo: Flat(10) Patient: Present
Temp: 30.00C Hum: 76.90% FSR: 873 Servo: Flat(10) Patient: Present
Temp: 30.00C Hum: 76.90% FSR: 848 Servo: Flat(10) Patient: Present
Temp: 30.00C Hum: 76.90% FSR: 742 Servo: Flat(10) Patient: Present
Temp: 30.00C Hum: 76.90% FSR: 0 Servo: Flat(10) Patient: Absent
Temp: 30.00C Hum: 76.90% FSR: 0 Servo: Flat(10) Patient: Absent
Temp: 30.00C Hum: 76.70% FSR: 0 Servo: Flat(10) Patient: Absent
```

Table 12.1 Observed Values from the Prototype Test Run

Parameter	Observed value / range	Condition
Temperature (DHT22)	30.0–30.1 °C	Across the run
Relative humidity (DHT22)	76.7–77.4 %	Across the run
FSR reading, patient present	742–873 (raw analog units)	Load on the bed
FSR reading, patient absent	0	Bed unoccupied
Servo position, tilted	Tilt, 120°	During tilt phase
Servo position, flat	Flat, 10°	After return-to-flat
Patient status flag	Present / Absent	Derived from FSR reading

12.1 Environmental Monitoring

Across the logged run, the DHT22 sensor reported a stable temperature of 30.0–30.1 °C and a relative humidity of 76.7–77.4 %, consistent with the indoor ambient conditions during testing. The small sample-to-sample variation indicates stable, low-noise environmental readings.

12.2 Pressure and Position Monitoring

While a patient/load was present on the bed, the FSR produced analog readings in the range of approximately 742–873 (raw sensor units); when the bed was unoccupied, the reading fell to 0. The firmware used this value to set the bed-occupancy status, reporting `Patient: Present` for a non-zero pressure reading and `Patient: Absent` once the reading dropped to 0. The pressure channel therefore reliably distinguished an occupied bed from an empty one.

12.3 Automatic Bed-Tilt Actuation

The servo-driven bed-tilt mechanism operated as intended. During the run the servo first held the bed in the tilted position (`Tilt(120)`, i.e. a 120° setting) while the patient was present, and then performed a return-to-flat cycle – logging the message `Returning Bed Flat (10 degrees)` and moving to the flat position (`Flat(10)`). This confirms that the automatic bed-tilt actuation is functional in the prototype.

12.4 Alert and Cloud Notification

This logging run captured the sensing and actuation status only; the buzzer, Wi-Fi mobile-notification, and cloud-upload events – and their delivery latency – were not recorded in this particular output. *[If you measured buzzer, notification, or cloud-upload timing separately, insert those figures here.]*

12.5 Discussion

The results confirm that the core subsystems of the prototype function together as intended: the DHT22 provided stable temperature and humidity readings, the FSR reliably distinguished an occupied bed from an empty one, and the servo actuator carried out the automatic tilt and return-to-flat cycle. Together these demonstrate a working end-to-end pathway from continuous sensing to automatic physical repositioning, which is the central objective of the system. A fuller quantitative characterization – pressure calibration against known reference loads, and buzzer, notification, and cloud-upload latency measurements – is identified as the next step in the evaluation.

CHAPTER 13

13. Advantages

- Reduces the risk of pressure ulcer development by providing continuous, automated tracking of position duration rather than relying solely on manually scheduled repositioning checks.
- Improves patient safety and comfort by combining pressure monitoring with environmental (temperature and humidity) sensing, addressing multiple contributing factors to skin breakdown in a single system.
- Reduces caregiver workload by automating the position-duration tracking task and by delivering both a local audible alert and a remote mobile notification, rather than requiring constant manual attention.
- Enables remote health monitoring, allowing doctors or caregivers to review a patient's pressure and environmental history from a cloud dashboard without needing to be physically present.
- Offers a low-cost, extensible smart-healthcare solution built from widely available IoT components, with an optional automatic bed-tilt feature available as a further, incremental extension.

14. Limitations

- The alerting subsystem depends on the availability of a stable Wi-Fi connection for mobile notifications and cloud data upload; local buzzer alerts remain functional without connectivity, but remote notification and cloud logging would be delayed or unavailable during an outage.
- FSR-based pressure sensing typically requires careful, patient-specific threshold calibration, since the relationship between applied force and sensor reading can vary with mattress type, sensor placement, and the patient's body weight.
- The DHT22 sensor's approximately two-second minimum sampling interval limits how quickly environmental changes can be detected, which is acceptable for slow-changing conditions such as ambient humidity but is not suited to rapid event detection.
- A discrete set of FSR sensor zones provides only point-based pressure sensing, and may not detect pressure changes occurring at body locations that fall between the instrumented zones.
- The optional automatic bed-tilt actuator adds mechanical complexity, cost, and safety considerations (such as ensuring gentle, fail-safe motion) that are not present in the base pressure-and-alert-only configuration.
- The system is designed as a monitoring and alerting aid and is not a substitute for clinical judgement, hands-on nursing care, or an established pressure-ulcer-prevention care plan.

CHAPTER 15

15. Future Work

While the design presented in this report establishes the core concept of an integrated, IoT-based patient-turning and bed-monitoring aid, several extensions have been identified as directions for future development beyond the scope of the initial prototype.

15.1 Multi-Zone, Full-Body Pressure Mapping

Future versions of the system could expand from a small number of discrete FSR zones to a denser sensor array or a textile pressure-mapping mattress topper, similar in spirit to the approach of Hudec et al. [2], providing a more complete picture of pressure distribution across the patient's body.

15.2 Machine-Learning-Based Risk Prediction

Historical pressure and environmental data logged to the cloud platform could be used to train a predictive model that flags patients at elevated pressure-ulcer risk before a threshold is actually crossed, enabling more proactive caregiver intervention.

15.3 Integration with Hospital Systems

A future version could integrate with hospital electronic health record (EHR) systems or existing nurse-call infrastructure, so that alerts generated by the device are logged directly against the patient's clinical record and routed through the facility's existing staff-notification workflow.

15.4 Wearable Vital-Sign Integration

Complementary wearable sensors – for heart rate, oxygen saturation, or additional motion sensing – could be integrated alongside the bed-based pressure and environmental sensors to provide a more complete picture of patient wellbeing.

15.5 Battery Backup and Power Resilience

A future revision could add a battery backup to the power-supply subsystem, ensuring that local buzzer alerting and sensor logging continue uninterrupted during a mains power outage.

CHAPTER 16

16. Project Cost

Table 16.1 presents an estimated bill of materials for a single working prototype of the IoT-Based Smart Patient Turning and Bed Monitoring System, based on representative retail component pricing from Bangladeshi electronics suppliers (TechShopBD, RoboticsBD, and BDTronics) at the time of writing. Costs are indicative and may vary by supplier, region, and order quantity; bulk purchasing for multiple units would be expected to reduce the per-unit cost further. The bill of materials reflects the configuration used for the testing prototype, in which a single FSR is used to demonstrate the core pressure-and-duration monitoring principle; the full multi-zone design described in Chapter 5 would use additional FSR units, one per monitored body zone. The table also includes the optional actuator components described in Chapter 5; a minimal, alert-only configuration (omitting the actuator) would cost less.

Item	Quantity	Unit Price (BDT)	Subtotal (BDT)
ESP32 DevKit V1	1	485	485
FSR Pressure Sensor (100 g–10 kg range)	1	820	820
DHT22 Temperature & Humidity Sensor	1	160	160
Active Buzzer (5 V)	1	15	15
LED Indicator (with resistor)	1	10	10
Servo Motor SG90 (optional actuator)	2	130	260
12 V Power Adapter	1	240	240
Wiring, Connectors & Miscellaneous	1 Set	300	300
Total (fully-featured configuration)			2,290

Table 16.1 Estimated Project Cost Analysis

Note: The IoT platform (Blynk or Firebase) is not included as a line item since both offer free tiers suitable for a single-prototype deployment. Prices are approximate, sourced from Bangladeshi electronics retailers, and are provided for indicative budgeting purposes only.

17. Conclusion

This report has presented the design, implementation, and evaluation of the IoT-Based Smart Patient Turning and Bed Monitoring System, an assistive device intended to reduce the risk of pressure ulcer development in bedridden patients through continuous, automated monitoring. The system combines pressure and position-duration sensing (using an FSR), environmental temperature and humidity monitoring (via a DHT22 sensor), local buzzer alerting, Wi-Fi-based mobile notification, cloud-based remote monitoring via the Blynk platform, and automatic bed-tilt actuation via a servo motor, all built around a low-cost ESP32 microcontroller. Together these address the three problems identified in Chapter 1: the difficulty of reliably tracking how long a patient has remained in one position, the lack of automated environmental monitoring, and the absence of a timely, automated caregiver-alerting pathway under the traditional manual-repositioning approach.

The literature reviewed in Chapter 2 confirms that the individual sensing and alerting capabilities underlying this work – pressure sensing, environmental sensing, mobile/cloud alerting, and automatic bed actuation – are each independently well established in prior work, while relatively few existing systems combine all of them into a single, low-cost, self-contained device. As reported in Chapter 12, the prototype was built and tested: it produced stable temperature and humidity readings, reliably distinguished an occupied bed from an empty one using the FSR, and carried out the automatic bed-tilt and return-to-flat cycle. With these core functions demonstrated, the system provides an affordable and extensible platform that can help reduce pressure-ulcer risk, improve patient safety and comfort, reduce caregiver workload, and support remote monitoring for doctors and caregivers, alongside the further enhancements discussed in Chapter 15.

18. References

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